

CSE3MAD

STEMM Lab

User Manual

A Real-World STEMM Games App

Assignment Phase 2

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1. Introduction

STEMM Lab is a mobile application that turns real-world physical activities into engaging, game-based learning experiences across Science, Technology, Engineering, Mathematics, and Medicine. The app is designed for upper primary and lower high school students working in teams.

For each challenge, students use their phone's sensors — the camera, microphone, accelerometer, timer and GPS — to capture real data, then review their results, rate the activity, and reflect on what they learned. This manual explains how to install and use every feature of the app.

What you will need

- An Android or iOS phone with the STEMM Lab app installed (APK build) or the Expo Go app.
- An internet connection to create an account and sync your data (the app also works offline).
- Everyday materials for each challenge, such as paper, string, a small toy and tape.



Figure 1.1 — The STEMM Lab splash screen on launch.

2. Getting Started — Creating an Account and Team

When you open STEMM Lab for the first time you are taken to the login screen. Each team signs in with its own account so that its data stays private and can be restored on any device.

2.1 Create an account

1. On the login screen, tap “Create an account”.
2. Enter a team email address and a password (at least six characters), then confirm the password.
3. Tap “Create Account”. Your account is created securely in the cloud.

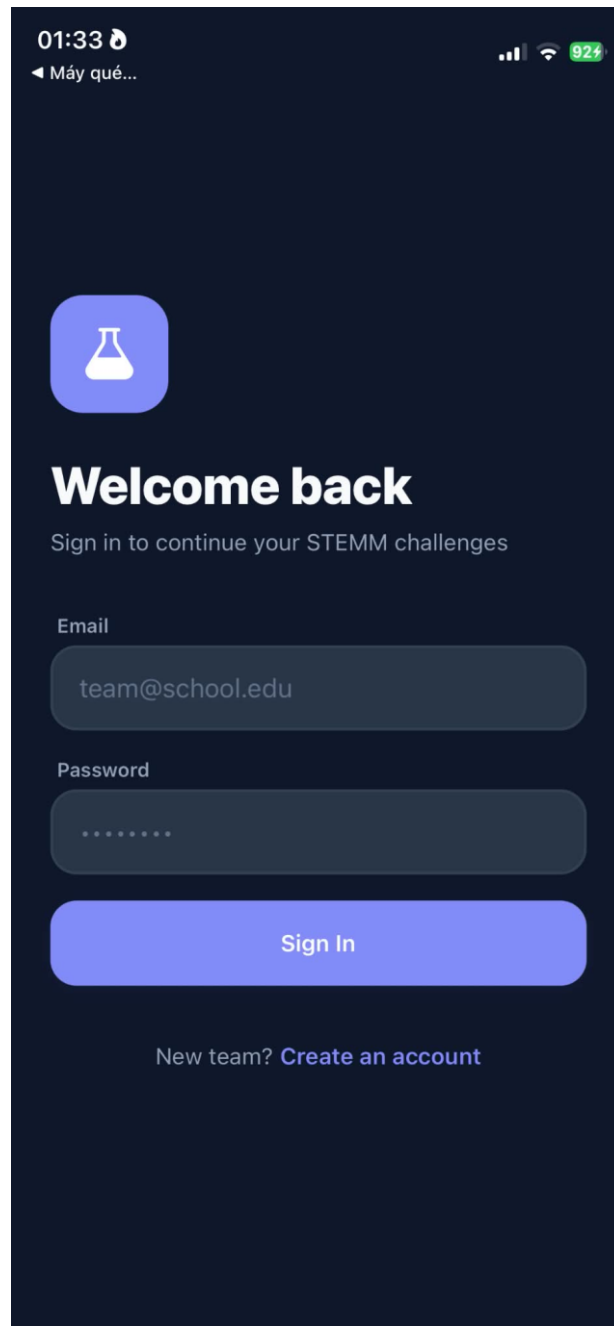


Figure 2.1 — The sign-up screen.

2.2 Set up your team

New accounts are taken straight to the team set-up screen. Returning teams skip this step and go directly to the home screen.

4. Enter your team name and your grade or year level.
5. Add the first name of each team member — tap “Add another member” for more.
6. Tap “Start Exploring”. The app assigns your team a unique four-digit code.

01:35

92%



Create account

Register your team to start exploring

Email

team@school.edu

Password

At least 6 characters

Confirm password

Re-enter password

Create Account

Already have an account? [Sign in](#)

Figure 2.2 — The team registration screen.

3. The Home Screen and Navigation

The home screen greets your team and displays all seven challenges in a two-column grid, split into Engineering Challenges and Health & Medical Sciences. Each card shows the activity icon, its category, and a status badge (Not started, In progress, or Completed).

Bottom navigation

- Home — the grid of all activities.
- Records — every saved result, with a QR code for sharing.
- Map — your completed activities pinned to where you recorded them.
- Team — your team profile, members and device battery.

The settings icon in the top-right of the home screen opens appearance and notification options.

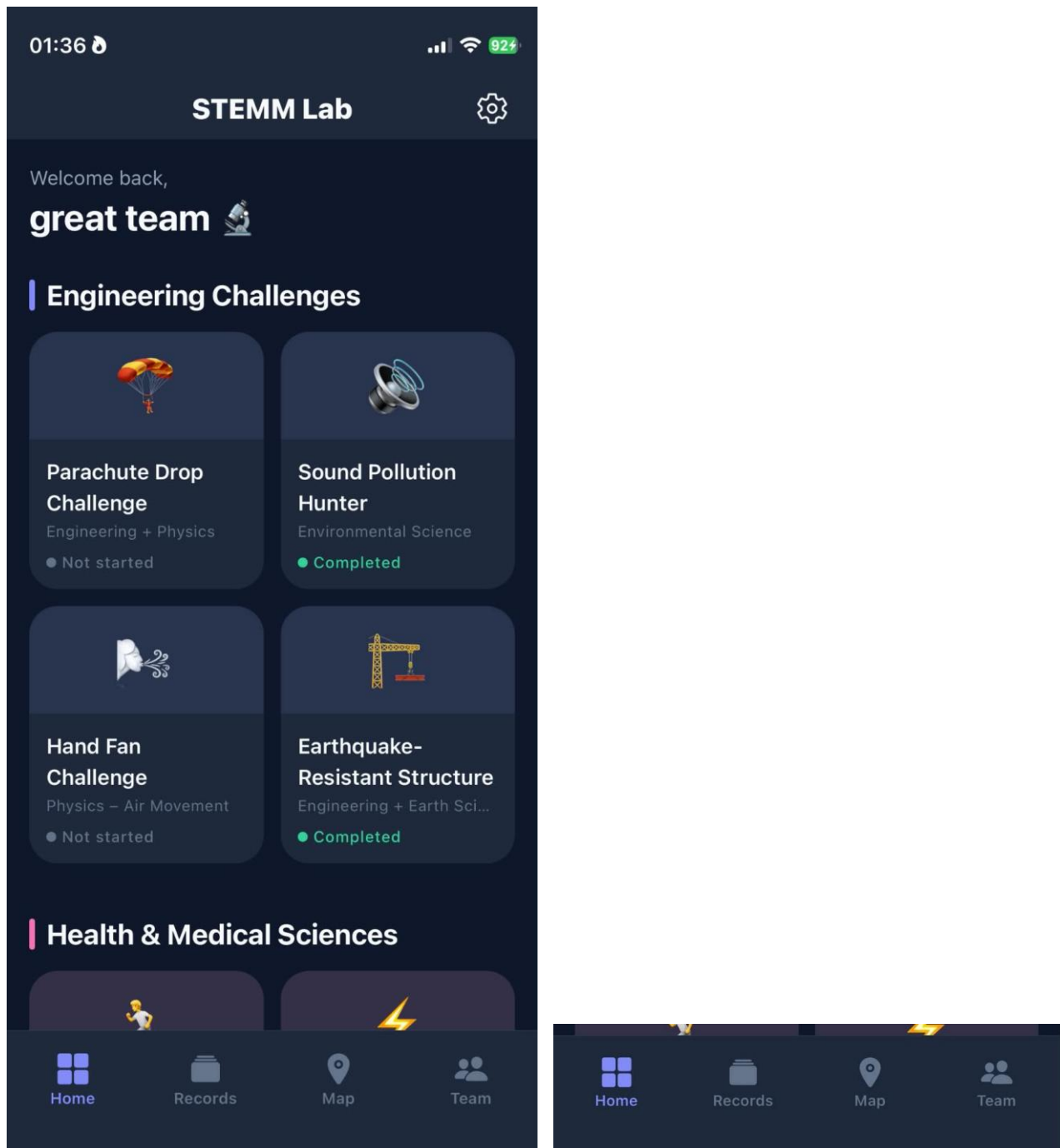


Figure 3.1 — The home screen with the activity grid and bottom tabs.

4. Completing an Activity

Every challenge follows the same four-step flow, so once you have completed one activity you will know how to complete them all.

7. Tap an activity card to open its Overview, which lists the equipment, the sensor used and what you will measure.
8. Tap “View Instructions” and follow the numbered steps. Steps that need a sensor are clearly marked.
9. Tap “Start Activity” to open the Record screen, where you capture live sensor data.
10. Tap “View Results” to see your calculated results, give the activity a star rating, write a team comment, then tap “Save & Finish”.

When you save, the result is stored on your device and synced to your account in the cloud, your location is tagged, and a “Activity complete” notification configuration

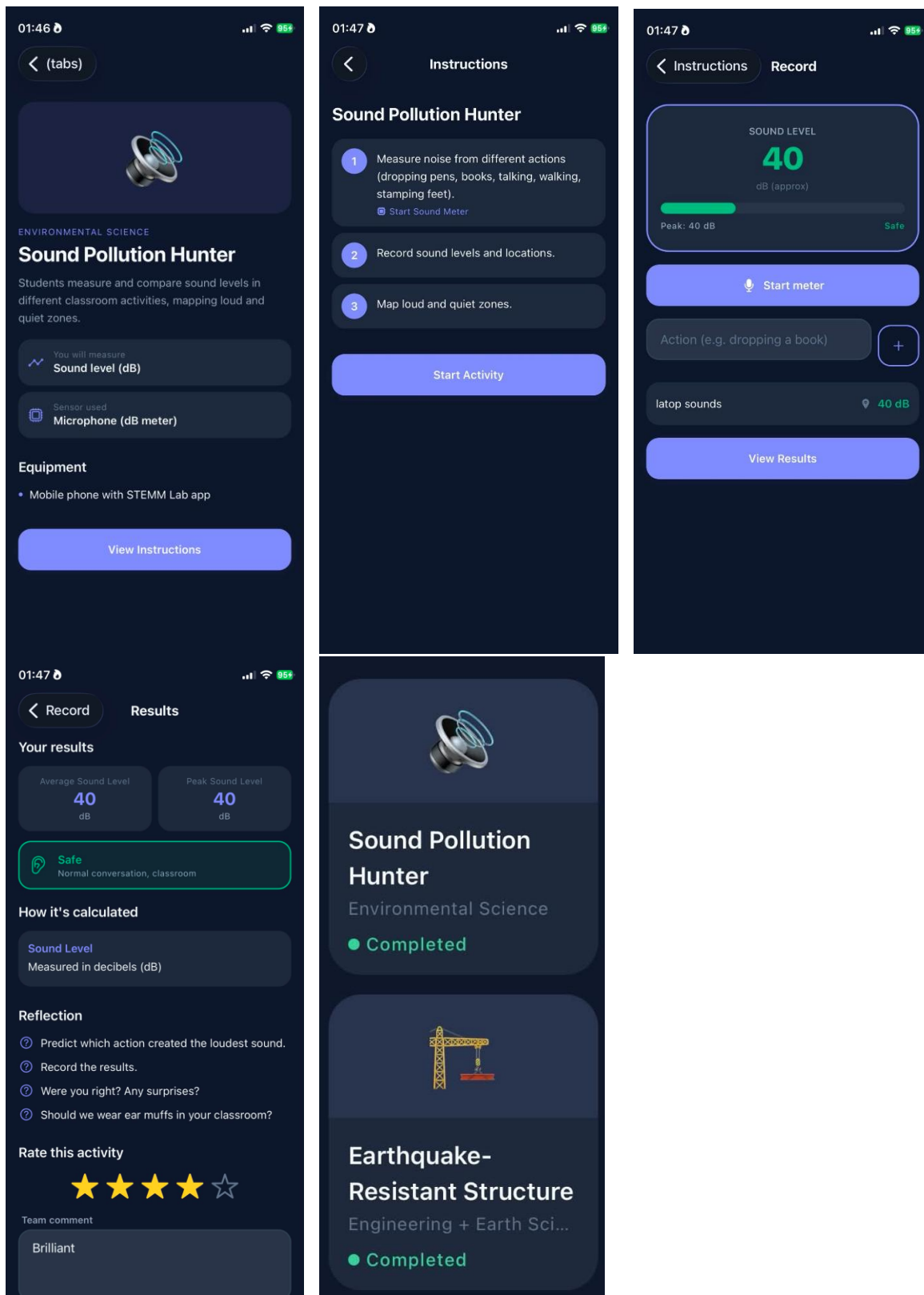


Figure 4.1 — The Overview, Instructions, Record and Results screens of sound pollution challenge.

The rest challenges are all the similar steps, so I just put 1 demonstration here! Please download file apk or repo code from my github to have full access and experience on my app.
<https://github.com/Chau05-git/STEMM-Lab-.git>

5. The Seven STEMM Activities

The table below summarises each challenge, the phone capability it uses, and what it measures.

Activity	Sensor / capability	What it measures
Parachute Drop	Camera + timer	Fall time, velocity, drag force and g-force
Sound Pollution Hunter	Microphone	Sound level in decibels and hearing-risk level
Hand Fan Challenge	Accelerometer	Bend angle and estimated force per material
Earthquake Structure	Accelerometer	Vibration (mm) — the steadiest design wins
Human Performance Lab	Accelerometer	Movement smoothness score and time
Reaction Board	Touchscreen	Reaction time in milliseconds
Breathing Pace Trainer	Accelerometer	Breaths per minute at rest and after exercise

5.1 Parachute Drop Challenge

Students test how parachute designs change falling speed. They drop a toy with and without a parachute, record the results, and compare which design lands slower and safer.

5.2 Sound Pollution Hunter

Students measure sound levels (dB) from different classroom actions. They compare noise levels, identify loud zones, and discuss how sound affects learning.

5.3 Hand Fan Challenge

Students use different fan designs to move a sheet of paper. They test how distance, stiffness, and fan shape affect the bending angle.

5.4 Earthquake-Resistant Structure

Students build simple structures and place a vibrating phone on top. They adjust folds and pillars to reduce movement and improve stability.

5.5 Human Performance Lab – Stretch Speed & Gracefulness

Students perform slow, controlled movements while holding a phone. The app measures vibration to show smoothness, speed, and coordination.

5.6 Reaction Board Challenge

Students test reaction time by tapping the screen, switching hands, and tracing moving shapes. They compare results across attempts and team members.

5.7 Breathing Pace Trainer

Students place the phone on their chest to measure breathing at rest and after exercise. They compare how breathing rate changes with activity.

6. Records

Records is where all the finished test reports are holding in the application. Users can go there and look all the challenges they have done so far including data, rating and comment on that challenge. Also the QR code is where student or teacher can take or transfer data to other without using external files or something. Just scan the code and user can see other reports !

6.1 Share a record

Each record has its own QR code. Open the record and show the QR code to another device.

6.2 Import a record

1. On the Records tab, tap “Scan a record QR”.
2. Point the camera at another device’s record QR code.
3. The record is imported and added to your account automatically.

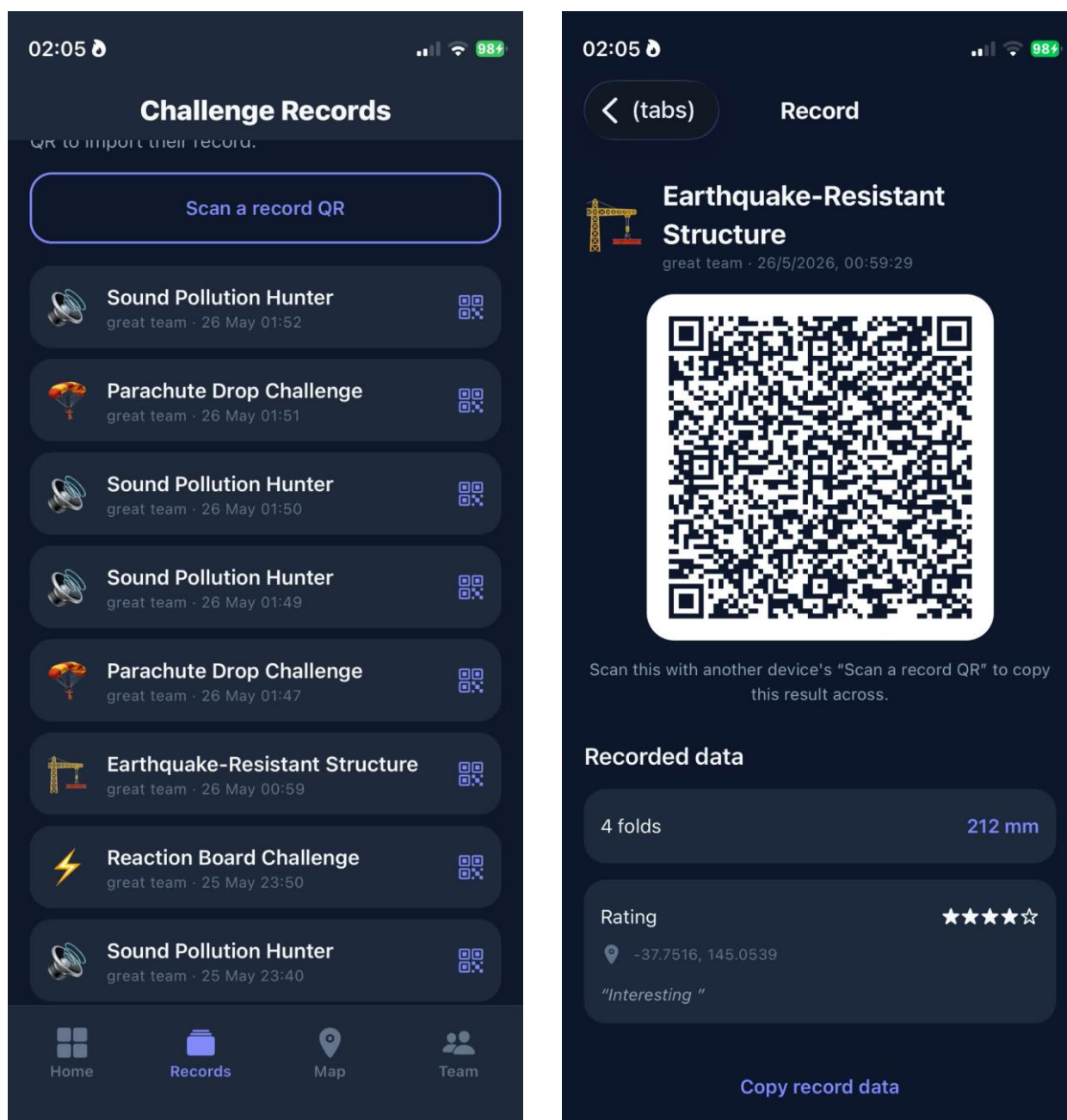


Figure 6.1 — Record screen and QR code.

7. The Activity Map

The Map tab shows your GPS-tagged activities as coloured pins — indigo for engineering challenges and pink for health challenges. Tap a pin to see the activity name, your team and the date it was recorded. A badge at the top counts how many activities have been tagged

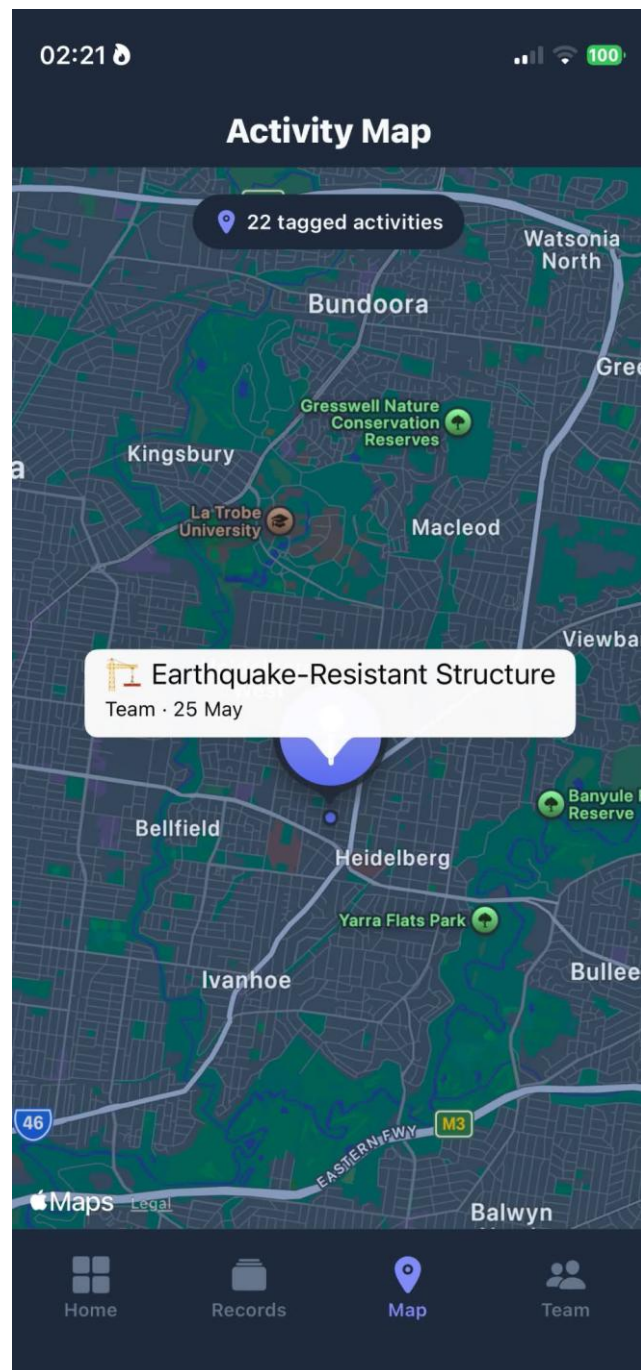


Figure 7.1 — The Map

8. Team Profile and Battery

The Team tab displays your team name, four-digit code, grade level and the list of members, along with how many activities you have completed. It also shows your device's live battery level, which turns amber and then red as the battery runs low. If the battery falls below 20% when you open an activity, the app warns you so a long recording is not interrupted.

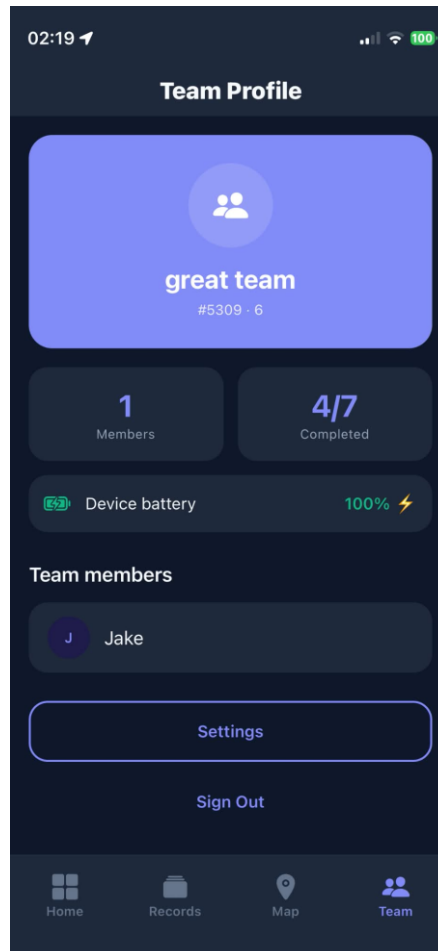


Figure 8.1 — The team profile and battery indicator.

9. Settings and Notifications

Open Settings from the home screen to change the appearance and preferences:

- Theme — choose System, Light or Dark.
- Notifications, Sound and Haptics — turn each on or off.

STEMM Lab sends helpful notifications: a confirmation when you complete an activity, a reminder if a challenge is left unfinished, and a warning when the battery is low.

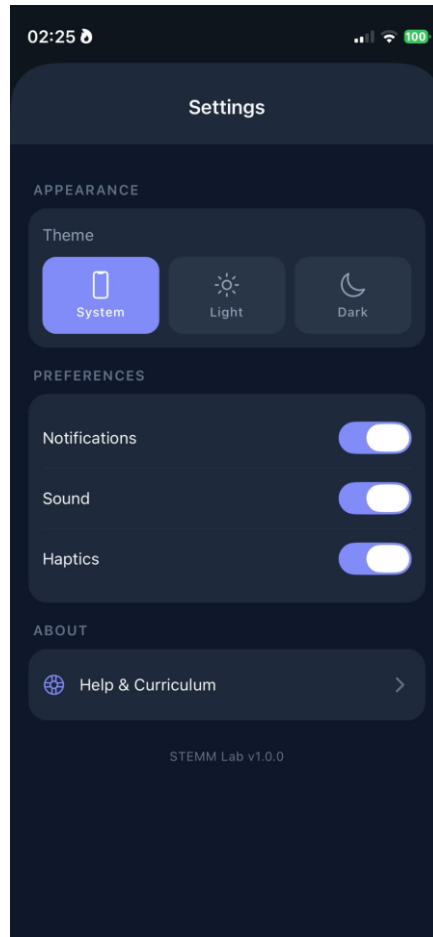


Figure 9.1 — The settings screen.

11. Troubleshooting

Problem	What to do
A result shows 0 or "--"	Make sure you started the sensor and saved a reading before viewing results.
No notification appears	Allow notifications when prompted, or enable them in Settings.
The map is empty	Complete an activity with location enabled; only tagged results appear.
My team is asked to register again	Sign in with the same account; your team is restored from the cloud.
Sensor permission denied	Grant camera, microphone and location permission in your phone settings.

End of User Manual.